



Brighter Star Drilling Fluids

ENGINEERED FOR HDD · MADE FOR RESULTS

HDD Drilling Fluid Product Manual

BENTONITE · CMC · MMH · CLAY VISCOSITY REDUCER · XANTHAN GUM

Laboratory testing, design

drilling fluid solutions

for your project specifically

Competitive Pricing

Guaranteed Quality

Ample Stock



BRSBENT SQ
(Premium Bentonite)



BRSCMC
(CMC)



BRSMMH
(Positive Electric Colloid)



BRSVR
(Clay Viscosity Reducer)



BRSXTG
(Xanthan Gum)

ABOUT US

A Team Built on Drilling Fluid Expertise

Our team brings **more than fifteen years of hands-on HDD construction experience**. Across Nigeria, China, Thailand, Pakistan and Bangladesh, we have completed **over 500 horizontal directional drilling crossings totalling more than 300 kilometres** — through silt, clay, sand, gravel, cobble and hard rock, with pipe diameters ranging from 6 inches all the way up to 56 inches.

Year after year on the job site taught us one lesson above all: **the correct use of drilling fluid is the foundation of every successful HDD crossing**. That is why we moved from using drilling fluids to engineering them. Drawing on our field record, we developed a complete family of HDD drilling fluid materials — bentonite, CMC, viscosity reducers, MMH and xanthan gum — each one formulated to solve the problems we know best, and each one verified in our own laboratory before it ever reaches a job site: viscosity profiles, fluid loss, carrying capacity and salt resistance are all measured on standard mud-testing equipment, so that the product you receive behaves exactly the way our data sheets promise.

15+

YEARS IN HDD

500+

HDD CROSSINGS

300+KILOMETRES
DRILLED**5**COUNTRIES
SERVED**6"–56"**

PIPE DIAMETERS



Our engineers running drilling fluid formulation and performance tests



Mud sample preparation for viscometer testing



Pouring a mud sample for Fann viscosity measurement



High-shear mixing of a bentonite-based mud formulation

every batch of Brighter Star products is validated on standard mud-testing equipment before delivery.

KNOW THE GROUND

HDD Challenges by Stratum

Each formation puts its own demands on the drilling fluid. Based on hundreds of crossings, these are the difficulties that decide whether a bore succeeds or fails — and each one is answered by a Brighter Star product in this manual.

Clay Formations

High viscosity

High torque

Pipe sticking / balling

Frac-out risk

Clay swells and sticks to the drill string and reamer, driving torque up sharply and choking the annulus. Without control, mud pressure builds until it breaks out to the surface.

Sandy Formations

Loose & unconsolidated

Borehole collapse

Pipe sticking

Frac-out risk

Sand offers no natural stability: the borehole wall caves in easily, cuttings settle fast, and the drill string can be buried. The fluid must build a tough, low-loss filter cake to hold the hole open.

Rock Formations

Poor cuttings transport

Secondary grinding

Severe tool wear

Very low ROP

Rock cuttings are heavy and angular. If the mud cannot carry them out, they are ground a second time at the bit — wearing out tooling and slowing the bore to a crawl.

Marine Sedimentary Formations

Mud system failure

Salt contamination

Pipe sticking

Salt water and high-salinity formations flocculate ordinary bentonite muds, destroying viscosity and filtration control precisely where the crossing is longest and hardest.

OUR ANSWER

The Brighter Star product family — BRSBENT SQ, BRSCMC, BRSM MH, BRSVR and BRSXTG — was engineered specifically against these four challenge profiles, and has been proven on more than 500 crossings worldwide.

Send us the geological survey report, the drawings, water samples (or water detection report). We will conduct the water testing and mud laboratory testing to provide a drilling fluid plan specifically for your project.

OUR PRODUCTS

The Brighter Star Product Family

Five materials cover the full range of HDD ground conditions. Used alone or in combination, they form a complete drilling fluid programme — from the base mud to the final pullback.



BRSBENT SQ

Premium bentonite — the foundation of every high-quality HDD mud



BRSCMC

Modified CMC, LV & HV grades — filtration control and viscosity



BRSM MH

Positive electric adhesive dry powder — superior carrying capacity



BRSVR

Clay viscosity reducer — cuts torque by up to 90%



BRSTG

Modified xanthan gum — stability in salt & marine formations



BRSBENT SQ — Premium Bentonite

THE FOUNDATION OF EVERY HIGH-QUALITY DRILLING FLUID

BRSBENT SQ is produced from ore mined in the high-grade montmorillonite belt of northern China, then professionally processed and treated with special additives. Bentonite is the base material of HDD drilling fluid — only top-quality bentonite can build a top-quality mud. Available in 25 kg bags and 1000 kg bulk bags.

- Outstanding viscosity build and suspension for the base mud
- Excellent filtration control and carrying capacity
- Suitable for all formation types



BRSCMC — Sodium Carboxymethyl Cellulose

FILTRATION CONTROL · VISCOSITY · BOREHOLE STABILITY

BRSCMC is based on Sodium Carboxymethyl Cellulose (CMC), specially modified into low-viscosity and high-viscosity grades, selectable according to your equipment capacity and the formation being crossed.

- Highly efficient fluid-loss reduction and viscosity building
- Greatly improves borehole stability in sandy formations
- Higher hole-making efficiency and a safer pullback

OUR PRODUCTS

Engineered Additives for Every Formation



BRSMMH — Positive Electric Adhesive Dry Powder

DYNAMIC SHEAR STRENGTH · CUTTINGS CARRYING IN ROCK

BRSMMH is a positive-electric adhesive dry powder developed with partner research institutes through extensive laboratory testing and field practice, formulated specifically for HDD applications.

- Greatly increases the dynamic shear (yield point) of the mud
- Exceptional cuttings-carrying capacity in rock crossings
- Field-proven on numerous highly difficult rock HDD projects



BRSVR — Clay Viscosity Reducer

CLAY DISPERSION · TORQUE REDUCTION UP TO 90%

BRSVR disperses drilled clay into the mud system instead of letting it agglomerate, sharply reducing its cohesion and viscosity. Clay balling on the drill string is greatly reduced — and with it the torque.

- Disperses clay and lowers mud viscosity in clay formations
- Dramatically reduces drill-pipe sticking and balling
- Torque reductions of over 90% recorded in field practice



BRSXTG — Modified Xanthan Gum

HIGH-SALINITY & MARINE SEDIMENTARY FORMATIONS

BRSXTG is a xanthan gum modified to suit HDD construction, ideal for high-salinity and seawater sedimentary formations. It keeps the mud system stable for extended periods so that key properties do not degrade — greatly reducing HDD risk in saline ground.

- Long-lasting mud stability under salt contamination
- Maintains viscosity and filtration performance in marine sediments
- Environmentally friendly, no hazard

SELECTION GUIDE

Stratum Characteristics & Product Applicability

Match the ground to the programme: what the ground does, what the mud must do, and the Brighter Star combination that has proven itself on our own crossings.

Clay

Ground: sticky, swelling cuttings; torque climbs; pipe balling and frac-out risk.

Mud: keep clay dispersed, viscosity low, annulus clean.

BRSBENT SQ + BRSVR

BRSVR disperses clay and cuts torque — over 90% in field records.

Sand & Gravel

Ground: unconsolidated, collapses easily; cuttings settle; high fluid loss.

Mud: tough filter cake, strong suspension, low filtration.

BRSBENT SQ + BRSCMC

Our most-used combination, from the Niger River to the Min Jiang.

Rock

Ground: heavy angular cuttings; secondary grinding; tool wear; low ROP.

Mud: high dynamic shear to lift cuttings out of the hole.

BRSBENT SQ + BRSMMH

BRSMMH lifts carrying capacity on long rock crossings such as Zhanjiang.

Marine Sedimentary / High Salinity

Ground: salt water flocculates standard muds; system failure; sticking.

Mud: salt-resistant system that holds its properties over time.

BRSBENT SQ + BRSXTG

BRSXTG keeps the system stable for the full duration of offshore crossings.

Table 1 Brighter Star Products — Function & Stratum Applicability

No.	Product	Main Ingredient	Increase Viscosity	Reduce Viscosity	Improve Dynamic Shear	Reduce Filtration	High-Salinity Resistance	Suitable Stratum	Env. Hazard
1	BRSBENT SQ	Premium bentonite	Excellent	N/A	Excellent	Excellent	N/A	All strata	None
2	BRSCMC	Sodium carboxymethyl cellulose	Very good	N/A	Very good	Excellent	N/A	Sand / gravel / rock	None
3	BRSMMH	Positive electric adhesive dry powder	Very good	N/A	Excellent	N/A	N/A	Rock	None
4	BRSVR	Clay viscosity reducer	N/A	Excellent	N/A	N/A	N/A	Clay	None
5	BRSXTG	Xanthan gum	Good	N/A	Good	Good	Excellent	High-salinity formations	None

PROVEN IN THE FIELD

Typical HDD Project Case Studies

Our products have been validated on a large number of demanding projects. The table below summarises eight representative crossings; detailed photos and product applications follow on the next pages.

Table 2 Typical HDD Projects Using Brighter Star Drilling Fluids

No.	Project	Country	Owner	Length	Dia.	Stratum	HDD Rig	Period	Products Used	Remark
1	OB3 RNC HDD Project	Nigeria	NGIC	2,000 m	48"	Sand / gravel	1200T / 500T	Nov.25–Apr.26	BRSBENT SQ + BRSCMC	Largest-scale HDD in Nigeria
2	AKK Niger River HDD Project	Nigeria	NGIC	1,560 m	40"	Sand / gravel / rock	1200T / 350T	Mar.25–Jul.25	BRSBENT SQ + BRSCMC	Niger River crossing
3	Fuzhou River HDD Crossing	China	SWG	2,100 m	40"	Sand	1000T / 600T	Feb.24–Jun.24	BRSBENT SQ + BRSCMC	EPC work
4	Zhanjiang Offshore HDD Crossing	China	CNOOC	2,200 m × 5	20"	Rock / sand (marine)	1000T / 600T	Aug.23–Apr.24	BRSBENT SQ + BRSMMH + BRSXTG	First 5-parallel onshore-to-offshore HDD
5	Raoyang River HDD Crossing	China	CNPC	2,293 m	48"	Sand / gravel / rock	1000T / 600T	Jun.22–Feb.23	BRSBENT SQ + BRSCMC	Among the largest HDD crossings in China
6	Tianjin LNG HDD Crossing	China	CNPC	655 / 750 / 860 m	56"	Clay / sand	1000T / 600T	Nov.20–Aug.21	BRSBENT SQ + BRSCMC / BRSVR	Largest pipe-size HDD crossing in China
7	5TP-1 36" Branch HDD Crossing	Thailand	PTT	1,680 / 1,700 m	36"	High-viscosity clay	600T	Jun.19–Jan.20	BRSBENT SQ + BRSVR	Longest large-diameter HDD in Thailand
8	5TP-1 HDD1 Gulf Crossing	Thailand	PTT	1,400 m	42"	Sand / granite rock	600T	Sep.19–Dec.19	BRSBENT SQ + BRSMMH	Largest-scale HDD in Thailand

WHY IT MATTERS TO YOU

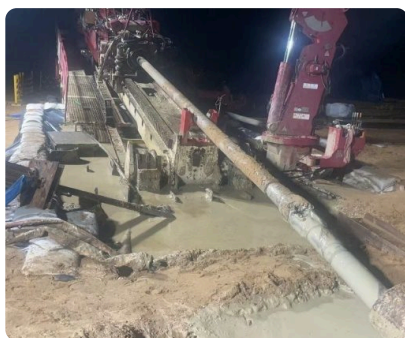
Rivers, seas, rock and clay — these products have already carried pipelines through the ground you are facing. When you choose Brighter Star, you are choosing formulations that have been proven on the largest and most difficult HDD crossings in Nigeria, China and Thailand.

OB3 RNC HDD Project

NIGERIA · NGIC

LENGTH × DIA. 2,000 m × 48"	DEPTH 52 m	STRATUM Sand / gravel	PRODUCTS BRSBENT SQ + BRSCMC
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A 2 km crossing of the Niger River at 52 m depth through highly complex sand and gravel strata — the largest-scale HDD project in Nigeria, completed with 1200T and 500T rigs working from both sides.



AKK Niger River HDD Project

NIGERIA · NGIC

LENGTH × DIA. 1,565 m × 40"	DEPTH 16 m	STRATUM Sand / gravel / rock	PRODUCTS BRSBENT SQ + BRSCMC
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A 1,565 m crossing of the Niger River through complex sand, gravel and rock — the second major Niger River crossing delivered with Brighter Star fluids on the AKK pipeline project.

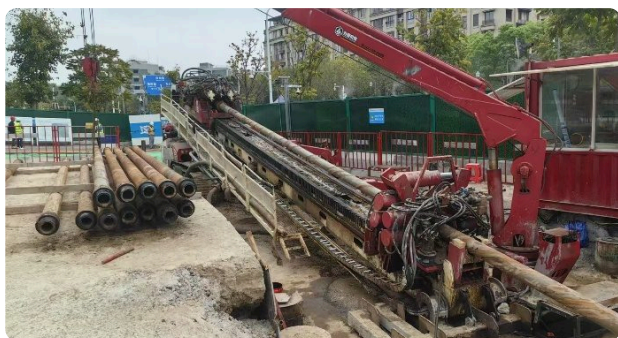


Fuzhou River HDD Crossing

CHINA · SWG

LENGTH × DIA. 2,100 m × 40"	DEPTH 35 m	STRATUM Sand / gravel	PRODUCTS BRSBENT SQ + BRSCMC
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A 2.1 km EPC crossing of the Min Jiang River at 35 m depth through complex sand and gravel — a showcase of filtration control and borehole stability with BRSBENT SQ + BRSCMC.

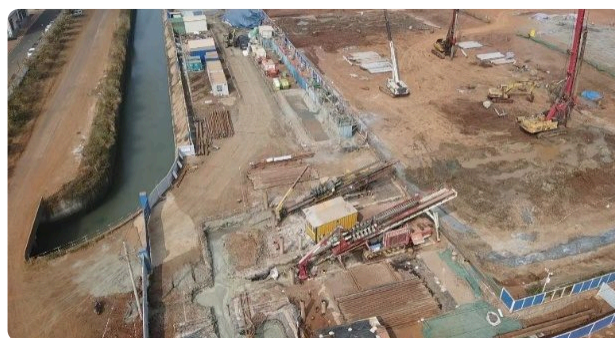
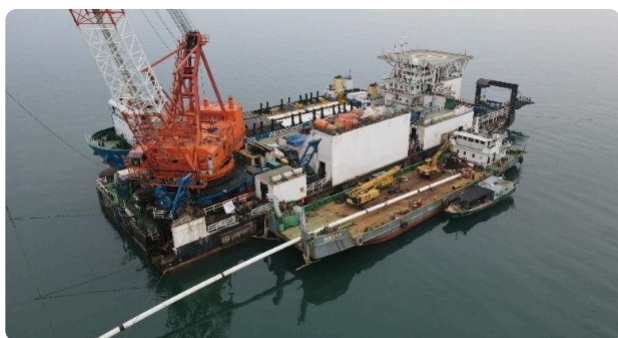


Zhanjiang Offshore HDD Crossing

CHINA · CNOOC

LENGTH × DIA. 2,200 m × 5 × 20"	DEPTH 42 m	STRATUM Rock / sand (marine)	PRODUCTS BRSBENT SQ + BRSMMH + BRSXTG
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Five parallel 2.2 km onshore-to-offshore crossings at 42 m depth through marine sedimentary rock and sand — the first project of its kind. BRSMMH carried the rock cuttings while BRSXTG kept the system stable under seawater contamination.

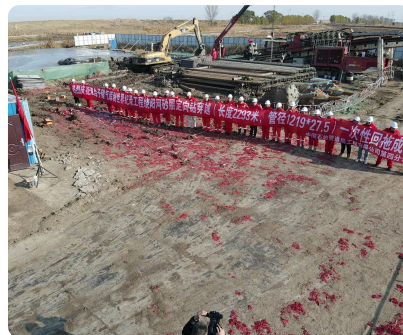


Raoyang River HDD Crossing

CHINA · CNPC

LENGTH × DIA. 2,293 m × 48"	DEPTH 27.5 m	STRATUM Sand / gravel / rock	PRODUCTS BRSBENT SQ + BRSCMC
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A 2.3 km, 48-inch crossing of the Raoyang River through sand, gravel and rock — among the largest HDD crossings of its kind in China.



Tianjin LNG HDD Crossing

CHINA · CNPC

LENGTH × DIA. 655 / 750 / 860 m × 56"	DEPTH 30 m	STRATUM Sand with clay	PRODUCTS BRSBENT SQ + BRSCMC / BRSVR
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Three crossings of local rivers with 56-inch pipe — the largest pipe-size HDD crossing in China — through sand with clay at 30 m depth.



5TP-1 36" Branch HDD Crossing

THAILAND · PTT

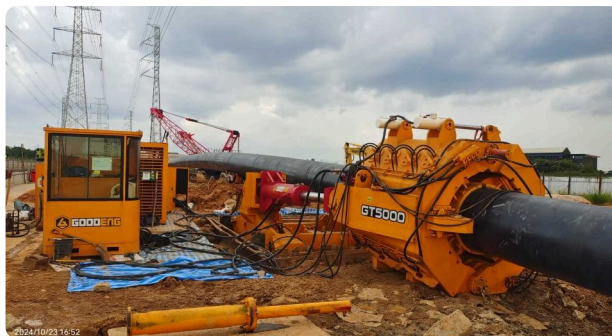
LENGTH × DIA.
1,680 / 1,700 m × 36"

DEPTH
20 m

STRATUM
High-viscosity clay

PRODUCTS
BRSBENT SQ + BRSVR

The longest large-diameter HDD in Thailand, through high-viscosity clay. BRSVR kept the clay dispersed and torque low across the full 1.7 km drive.



5TP-1 HDD1 Gulf Crossing

THAILAND · PTT

LENGTH × DIA.
1,400 m × 42"

DEPTH
20 m

STRATUM
Sand / granite rock

PRODUCTS
BRSBENT SQ +
BRSMH

The largest-scale HDD in Thailand — 1.4 km through sand and granite. BRSMH gave the mud the carrying capacity needed to keep granite cuttings moving to surface.



READY WHEN YOU ARE

Ample Stock in Nigeria

Brighter Star maintains a **large standing inventory of drilling fluid materials inside Nigeria**. Whatever your programme — bentonite, CMC, MMH, clay viscosity reducer or xanthan gum — the products you need are already in-country, palletised and ready for immediate dispatch to your site.



Bulk-bag bentonite inventory, Nigeria warehouse



1000 kg big bags stacked and ready for dispatch



25 kg bagged products on pallets



Mixed product inventory held in-country

NO WAITING ON SHIPMENTS

Because the stock is already in Nigeria, your order moves from warehouse to rig site in days, not months — keeping your drilling programme on schedule.



Brighter Star Drilling Fluids

COMPETITIVE PRICING · GUARANTEED QUALITY · AMPLE STOCK · THE
FIRST CHOICE FOR HDD

www.pcenigeria.com

When the ground gets difficult, choose the products that have already crossed it.

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